AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M140
Introducing Statistics

Final Examination
Fall Semester 2025/2026

Date: 22/Jan./2026

Number of Exam Pages: (3) (including this cover sheet)	**Time Allowed: 3 Hours**

Instructions:

Please read the following instructions before starting:

- Total Number of Questions: 5
- This is a closed book exam.
- Be sure you write your name and ID on the External Answer booklet.
- This exam accounts to 50% of the total mark in the course.
- Budget your time according to the mark assigned to each question.
- Mobile phones and all other mobile communication equipment are N allowed.
- Calculators are allowed.

CamScanner

Part 1: MULTIPLE CHOICE QUESTIONS

Solve all the questions. Each question is worth 4 marks, giving a total of 20 possible marks.

Q–1: Which of the following can't be the probability of an event?

a) 0 c) -4/7

b) 0.5 d) 1

Q–2: Nationality is an example of what level of measurement?

a) ordinal c) ratio

b) nominal d) interval

Q–3: If a variable assumes the values 2, 3, 5, and 8, then this variable is a ..... variable.

a) continuous c) discrete

b) continuous and quantitative d) discrete and qualitative

Q–4: What is the statistical term for the mean that is obtained by using the data values of a sample drawn from a specific population

a) parameter c) variable

b) statistic d) quantity

Q–5: The standard normal distribution is .....

a) skewed to the right c) symmetric

b) skewed to the left d) asymmetric

Part 2: ESSAY QUESTIONS

Each question is worth 20 marks. You should answer all FOUR questions, giving a total of 80 possible marks.

Q–1: [10+10 Marks]

a) consider the following sets of data:

A: 3, 5, 7, 9, 11 ; B: 3, 7, 7, 7, 11

(I) Find the standard deviation of each set of data

(II) Which of the two sets of data is more dispersed?

b) The following table gives 400 patients classified by sex and blood group.

		Male (M)	Female (F)	Total
Blood group	O	100	74	174
	A	90	66	156
	B	25	18	43
	AB	15	12	27
Total		230	170	400

If a patient is selected at random from this group, find the probabilities of the following events:

(i) $A \cap M$ (ii) $M \cap F$ (iii) $M \cup F$

CamScanner

Q-2: [20 Marks]

To check the existence of a relationship between smoking and having lung cancer, a sample of 100 was chosen and give the following table:

	Having lung cancer	Normal
Heavy smoker	14	6
ex-smoking	11	10
Never smoke	5	54

We want to use the sample data to decide whether there is an association between smoking and having lung cancer at 0.05 level of significance.
[Given: $\chi^2_{0.05,2} = 5.991$]

Q–3: [10+10 Marks]

a) The waiting times (in minutes) of patients who attend a certain clinic have a normal distribution with an unknown mean μ and standard deviation 4 minutes. If the mean waiting time of 16 patients selected at random is 15 minutes. Find a 90% confidence interval for μ.

[Given: $z_{0.05} = 1.645$]

b) The heights of students of a certain school are normally distributed with mean 160 cm and standard deviation 8 cm. What proportion of students have the heights between 152 cm and 176 cm ?

Hint: Use the table below

z	-1	-2.00	1.17	1.18	1.60	2.00	1.80	3.00
P (Z < z)	0.1587	0.0228	0.8790	0.8810	0.9452	0.9772	0.9641	0.9987

Q–4: [20 Marks]

An electrical firm manufactures light bulbs that have a lifetime that is approximately normally distributed with a mean of 800 hours and a standard deviation of 40 hours. Test the hypothesis that $\mu = 800$ hours against the alternative $\mu \neq 800$ hours if a random sample of 30 bulbs has an average life of 788 hours. the 0.05 level of significance.
[Given: $Z_{0.025} = 1.96$]

End of Questions

CamScanner